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Inventor(s)

Michaels; Andrew Steil et al.

LIDAR SENSOR SYSTEM

Abstract

A light detection and ranging (LIDAR) sensor system for a vehicle includes a laser source, one or more optics, and one or more scanners. The laser source is configured to generate a beam. The one or more optics are configured to provide a modified beam by modifying a beam shape of the beam to be outside of a boundary of a target shape for the beam. The one or more scanners are configured to output the modified beam.

Inventors: Michaels; Andrew Steil (Los Altos, CA), Rogers; Evan (Bozeman, MT)

Applicant: Aurora Operations, Inc. (Pittsburgh, PA)

Family ID: 96881229

Assignee: Aurora Operations, Inc. (Pittsburgh, PA)

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Background/Summary

BACKGROUND

[0001] Optical detection of range using lasers, often referenced by a mnemonic, LIDAR (for “light detection and ranging”), also sometimes referred to as “laser RADAR,” is used for a variety of applications, including imaging and collision avoidance. LIDAR provides finer scale range resolution with smaller beam sizes than conventional microwave ranging systems, such as radio-wave detection and ranging (RADAR).

SUMMARY

[0002] At least one aspect relates to a light detection and ranging (LIDAR) system, such as a LIDAR sensor system for a vehicle. The LIDAR sensor system includes a laser source, one or more optics, and one or more scanners. The laser source is configured to generate a beam. The one or more optics are configured to provide a modified beam by modifying a beam shape of the beam to be outside of a boundary of a target shape for the beam. The one or more scanners are configured to output the modified beam.

[0003] At least one aspect relates to an autonomous vehicle control system. The autonomous vehicle control system includes a laser source, one or more optics, one or more scanners, and one or more processors. The laser source is configured to generate a beam. The one or more optics are configured to provide a modified beam by modifying a beam shape of the beam to be outside of a boundary of a target shape for the beam. The one or more scanners are configured to output the modified beam. The one or more processors are configured to determine at least one of a range to an object or a velocity of the object based on a return signal from reflection of the modified beam by the object, and to control operation of an autonomous vehicle responsive to the at least one of the range or the velocity.

[0004] At least one aspect relates to an autonomous vehicle. The autonomous vehicle includes a LIDAR sensor system, a steering system, a braking system, and a vehicle controller. The LIDAR sensor system includes a laser source, one or more optics, and one or more scanners. The laser source is configured to generate a beam. The one or more optics are configured to provide a modified beam by modifying a beam shape of the beam to be outside of a boundary of a target shape for the beam. The one or more scanners are configured to output the modified beam. The vehicle controller includes one or more processors configured to determine at least one of a range to an object or a velocity of the object based on a return signal from reflection of the modified beam by the object, and to control operation of at least one of the steering system and the braking system responsive to the at least one of the range or the velocity.

[0005] Those skilled in the art will appreciate that the summary is illustrative only and is not intended to be in any way limiting. Any of the features described herein may be used with any other features, and any subset of such features can be used in combination according to various embodiments. Other aspects, inventive features, and advantages of the devices and/or processes described herein, as defined solely by the claims, will become apparent in the detailed description set forth herein and taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] Implementations are illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings in which like reference numerals refer to similar elements and in which:

[0007] FIG. 1 is a block diagram of an example of a system environment for autonomous vehicles;

[0008] FIG. 2 is a block diagram of an example of a system environment for autonomous commercial trucking vehicles;

[0009] FIG. 3 is a block diagram of an example of a system environment for autonomous

commercial trucking vehicles;

[0010] FIG. **4** is a block diagram of an example of a system environment for autonomous commercial trucking vehicles;

[0011] FIG. **5** is a block diagram of an example of a LIDAR sensor system;

[0012] FIG. **6** is a block diagram of an example of an optic module of a LIDAR sensor system;

[0013] FIG. **7** is a block diagram of an example of a LIDAR sensor system;

[0014] FIG. **8** is a block diagram of an example of optical components of a system;

[0015] FIG. **9** is a block diagram of an example of a LIDAR sensor system including the optical components of FIG. **8**;

[0016] FIG. **10** is a block diagram of an example of optical components of a system;

[0017] FIG. **11** is a block diagram of an example of a LIDAR sensor system including the optical components of FIG. **10**;

[0018] FIG. **12** is a block diagram of an example of optical components of a system;

[0019] FIG. **13** is a block diagram of an example of optical components of a system;

[0020] FIG. **14** is a block diagram of an example of optical components of a system;

[0021] FIG. **15** is a block diagram of an example of a LIDAR sensor system including the optical components of FIG. **14**;

[0022] FIG. **16** is a block diagram of an example of optical components of a system;

[0023] FIG. **17** is a block diagram of an example of a circulator; and

[0024] FIG. **18** is a block diagram of an example of a LIDAR sensor system that includes a beam shape optic.

DETAILED DESCRIPTION

[0025] A LIDAR sensor system can generate and transmit a light beam that an object can reflect or otherwise scatter as a return beam corresponding to the transmitted beam. The LIDAR sensor system can receive the return beam, and process the return beam or characteristics thereof to determine parameters regarding the object such as range and velocity. The LIDAR sensor system can apply various frequency or phase modulations to the transmitted beam, which can facilitate relating the return beam to the transmitted beam in order to determine the parameters regarding the object.

[0026] Systems and methods in accordance with the present disclosure can allow for LIDAR sensor systems to more effectively detect objects having low likelihood of reflections being returned to the LIDAR sensors systems, such as due to the objects having dark paint (e.g., dark automotive paint on vehicles). For example, many of the points on such objects that the transmit beams transmitted by the LIDAR sensor system hit, other than tail lights or license plates, may not result (or infrequently result) in return beams being reflected to the LIDAR sensor system.

[0027] For example, one or more optics of the LIDAR sensor system, such as a collimator and/or collimator lens, can be structured to allow for a controlled amount of beam blooming (e.g., extension or redirection of light and/or power outside of a collimated, diffraction-limited, and/or Gaussian shape for the beam) in order to more effectively detect specular reflections. The one or more optics can implement blooming in a manner so that the LIDAR sensor system can detect points of an object within an angular resolution distance of the beam, even if the beam does not directly fall on the points, allowing for more complete detection of dark objects. This can allow for more effective processes to be performed on the LIDAR data, including but not limited to object detection and/or perception tasks used for controlling an autonomous vehicle. However, the advantages of the optics described above are not limited to autonomous vehicles. They can be advantageous for any type of vehicles equipped with LIDAR sensors.

1. System Environments for Autonomous Vehicles

[0028] FIG. **1** is a block diagram illustrating an example of a system environment for autonomous vehicles according to some implementations. FIG. **1** depicts an example autonomous vehicle **100** within which the various techniques disclosed herein may be implemented. The vehicle **100**, for

example, may include a powertrain **102** including a prime mover **104** powered by an energy source **106** and capable of providing power to a drivetrain **108**, as well as a control system **110** including a direction control **112**, a powertrain control **114**, and a brake control **116**. The vehicle **100** may be implemented as any number of different types of vehicles, including vehicles capable of transporting people and/or cargo, and capable of traveling in various environments. The aforementioned components **102-116** can vary widely based upon the type of vehicle within which these components are utilized, such as a wheeled land vehicle such as a car, van, truck, or bus. The prime mover **104** may include one or more electric motors and/or an internal combustion engine (among others). The energy source may include, for example, a fuel system (e.g., providing gasoline, diesel, hydrogen, etc.), a battery system, solar panels or other renewable energy source, and/or a fuel cell system. The drivetrain **108** can include wheels and/or tires along with a transmission and/or any other mechanical drive components to convert the output of the prime mover **104** into vehicular motion, as well as one or more brakes configured to controllably stop or slow the vehicle **100** and direction or steering components suitable for controlling the trajectory of the vehicle **100** (e.g., a rack and pinion steering linkage enabling one or more wheels of the vehicle **100** to pivot about a generally vertical axis to vary an angle of the rotational planes of the wheels relative to the longitudinal axis of the vehicle). In some implementations, combinations of powertrains and energy sources may be used (e.g., in the case of electric/gas hybrid vehicles), and in some instances multiple electric motors (e.g., dedicated to individual wheels or axles) may be used as a prime mover.

[0029] The direction control **112** may include one or more actuators and/or sensors for controlling and receiving feedback from the direction or steering components to enable the vehicle **100** to follow a desired trajectory. The powertrain control **114** may be configured to control the output of the powertrain **102**, e.g., to control the output power of the prime mover **104**, to control a gear of a transmission in the drivetrain **108**, etc., thereby controlling a speed and/or direction of the vehicle **100**. The brake control **116** may be configured to control one or more brakes that slow or stop vehicle **100**, e.g., disk or drum brakes coupled to the wheels of the vehicle.

[0030] Other vehicle types, including but not limited to off-road vehicles, all-terrain or tracked vehicles, construction equipment, may utilize different powertrains, drivetrains, energy sources, direction controls, powertrain controls and brake controls. Moreover, in some implementations, some of the components can be combined, e.g., where directional control of a vehicle is primarily handled by varying an output of one or more prime movers.

[0031] Various levels of autonomous control over the vehicle **100** can be implemented in a vehicle control system **120**, which may include one or more processors **122** and one or more memories **124**, with each processor **122** configured to execute program code instructions **126** stored in a memory **124**. The processors(s) can include, for example, graphics processing unit(s) (“GPU(s)”) and/or central processing unit(s) (“CPU(s)”).

[0032] Sensors **130** may include various sensors suitable for collecting information from a vehicle's surrounding environment for use in controlling the operation of the vehicle. For example, sensors **130** can include radar sensor **134**, LIDAR (Light Detection and Ranging) sensor **136**, a 3D positioning sensors **138**, e.g., any of an accelerometer, a gyroscope, a magnetometer, or a satellite navigation system such as GPS (Global Positioning System), GLONASS (Globalnaya Navigazionnaya Sputnikovaya Sistema, or Global Navigation Satellite System), BeiDou Navigation Satellite System (BDS), Galileo, Compass, etc. The 3D positioning sensors **138** can be used to determine the location of the vehicle on the Earth using satellite signals. The sensors **130** can include a camera **140** and/or an IMU (inertial measurement unit) **142**. The camera **140** can be a monographic or stereographic camera and can record still and/or video images. The IMU **142** can include multiple gyroscopes and accelerometers capable of detecting linear and rotational motion of the vehicle in three directions. One or more encoders (not illustrated), such as wheel encoders may be used to monitor the rotation of one or more wheels of vehicle **100**. Each sensor **130** can

output sensor data at various data rates, which may be different than the data rates of other sensors **130**.

[0033] The outputs of sensors **130** may be provided to a set of control subsystems **150**, including a localization subsystem **152**, a planning subsystem **156**, a perception subsystem **154**, and a control subsystem **158**. The localization subsystem **152** can perform functions such as precisely determining the location and orientation (also sometimes referred to as “pose”) of the vehicle **100** within its surrounding environment, and generally within some frame of reference. The location of an autonomous vehicle can be compared with the location of an additional vehicle in the same environment as part of generating labeled autonomous vehicle data. The perception subsystem **154** can perform functions such as detecting, tracking, determining, and/or identifying objects within the environment surrounding vehicle **100**. A machine learning model in accordance with some implementations can be utilized in tracking objects. The planning subsystem **156** can perform functions such as planning a trajectory for vehicle **100** over some timeframe given a desired destination as well as the static and moving objects within the environment. A machine learning model in accordance with some implementations can be utilized in planning a vehicle trajectory. The control subsystem **158** can perform functions such as generating suitable control signals for controlling the various controls in the vehicle control system **120** in order to implement the planned trajectory of the vehicle **100**. A machine learning model can be utilized to generate one or more signals to control an autonomous vehicle to implement the planned trajectory.

[0034] Multiple sensors of types illustrated in FIG. **1** can be used for redundancy and/or to cover different regions around a vehicle, and other types of sensors may be used. Various types and/or combinations of control subsystems may be used. Some or all of the functionality of a subsystem **152-158** may be implemented with program code instructions **126** resident in one or more memories **124** and executed by one or more processors **122**, and these subsystems **152-158** may in some instances be implemented using the same processor(s) and/or memory. Subsystems may be implemented at least in part using various dedicated circuit logic, various processors, various field programmable gate arrays (“FPGA”), various application-specific integrated circuits (“ASIC”), various real time controllers, and the like, as noted above, multiple subsystems may utilize circuitry, processors, sensors, and/or other components. Further, the various components in the vehicle control system **120** may be networked in various manners.

[0035] In some implementations, the vehicle **100** may also include a secondary vehicle control system (not illustrated), which may be used as a redundant or backup control system for the vehicle **100**. In some implementations, the secondary vehicle control system may be capable of fully operating the autonomous vehicle **100** in the event of an adverse event in the vehicle control system **120**, while in other implementations, the secondary vehicle control system may only have limited functionality, e.g., to perform a controlled stop of the vehicle **100** in response to an adverse event detected in the primary vehicle control system **120**. In still other implementations, the secondary vehicle control system may be omitted.

[0036] Various architectures, including various combinations of software, hardware, circuit logic, sensors, and networks, may be used to implement the various components illustrated in FIG. **1**. Each processor may be implemented, for example, as a microprocessor and each memory may represent the random access memory (“RAM”) devices comprising a main storage, as well as any supplemental levels of memory, e.g., cache memories, non-volatile or backup memories (e.g., programmable or flash memories), read-only memories, etc. In addition, each memory may be considered to include memory storage physically located elsewhere in the vehicle **100**, e.g., any cache memory in a processor, as well as any storage capacity used as a virtual memory, e.g., as stored on a mass storage device or another computer controller. One or more processors illustrated in FIG. **1**, or entirely separate processors, may be used to implement additional functionality in the vehicle **100** outside of the purposes of autonomous control, e.g., to control entertainment systems, to operate doors, lights, convenience features, etc.

[0037] In addition, for additional storage, the vehicle **100** may include one or more mass storage devices, e.g., a removable disk drive, a hard disk drive, a direct access storage device (“DASD”), an optical drive (e.g., a CD drive, a DVD drive, etc.), a solid state storage drive (“SSD”), network attached storage, a storage area network, and/or a tape drive, among others.

[0038] Furthermore, the vehicle **100** may include a user interface **164** to enable vehicle **100** to receive a number of inputs from and generate outputs for a user or operator, e.g., one or more displays, touchscreens, voice and/or gesture interfaces, buttons and other tactile controls, etc. Otherwise, user input may be received via another computer or electronic device, e.g., via an app on a mobile device or via a web interface.

[0039] Moreover, the vehicle **100** may include one or more network interfaces, e.g., network interface **162**, suitable for communicating with one or more networks **170** (e.g., a Local Area Network (“LAN”), a wide area network (“WAN”), a wireless network, and/or the Internet, among others) to permit the communication of information with other computers and electronic device, including, for example, a central service, such as a cloud service, from which the vehicle **100** receives environmental and other data for use in autonomous control thereof. Data collected by the one or more sensors **130** can be uploaded to a computing system **172** via the network **170** for additional processing. In some implementations, a time stamp can be added to each instance of vehicle data prior to uploading.

[0040] Each processor illustrated in FIG. 1, as well as various additional controllers and subsystems disclosed herein, generally operates under the control of an operating system and executes or otherwise relies upon various computer software applications, components, programs, objects, modules, data structures, etc., as will be described in greater detail below. Moreover, various applications, components, programs, objects, modules, etc. may also execute on one or more processors in another computer coupled to vehicle **100** via network **170**, e.g., in a distributed, cloud-based, or client-server computing environment, whereby the processing required to implement the functions of a computer program may be allocated to multiple computers and/or services over a network.

[0041] In general, the routines executed to implement the various implementations described herein, whether implemented as part of an operating system or a specific application, component, program, object, module or sequence of instructions, or even a subset thereof, will be referred to herein as “program code”. Program code can include one or more instructions that are resident at various times in various memory and storage devices, and that, when read and executed by one or more processors, perform the steps necessary to execute steps or elements embodying the various aspects of the present disclosure. Moreover, while implementations have and hereinafter will be described in the context of fully functioning computers and systems, it will be appreciated that the various implementations described herein are capable of being distributed as a program product in a variety of forms, and that implementations can be implemented regardless of the particular type of computer readable media used to actually carry out the distribution.

[0042] Examples of computer readable media include tangible, non-transitory media such as volatile and non-volatile memory devices, floppy and other removable disks, solid state drives, hard disk drives, magnetic tape, and optical disks (e.g., CD-ROMs, DVDs, etc.) among others.

[0043] In addition, various program code described hereinafter may be identified based upon the application within which it is implemented in a specific implementation. Any particular program nomenclature that follows is used merely for convenience, and thus the present disclosure should not be limited to use solely in any specific application identified and/or implied by such nomenclature. Furthermore, given the typically endless number of manners in which computer programs may be organized into routines, procedures, methods, modules, objects, and the like, as well as the various manners in which program functionality may be allocated among various software layers that are resident within a typical computer (e.g., operating systems, libraries, API's, applications, applets, etc.), the present disclosure is not limited to the specific organization and

allocation of program functionality described herein.

2. LIDAR for Automotive Applications

[0044] A truck can include a LIDAR system (e.g., vehicle control system **120** in FIG. **1**, LIDAR sensor system **500** in FIG. **5**, among others described herein). In some implementations, the LIDAR sensor system **500** can use frequency modulation to encode an optical signal and scatter the encoded optical signal into free-space using optics. By detecting the frequency differences between the encoded optical signal and a returned signal reflected back from an object, the frequency modulated (FM) LIDAR sensor system can determine the location of the object and/or precisely measure the velocity of the object using the Doppler effect. In some implementations, an FM LIDAR sensor system may use a continuous wave (referred to as, “FMCW LIDAR”) or a quasi-continuous wave (referred to as, “FMQW LIDAR”). In some implementations, the LIDAR sensor system can use phase modulation (PM) to encode an optical signal and scatters the encoded optical signal into free-space using optics.

[0045] In some instances, an object (e.g., a pedestrian wearing dark clothing) may have a low reflectivity, in that it only reflects back to the sensors (e.g., sensors **130** in FIG. **1**) of the FM or PM LIDAR sensor system a low amount (e.g., 10% or less) of the light that hit the object. In other instances, an object (e.g., a shiny road sign) may have a high reflectivity (e.g., above 10%), in that it reflects back to the sensors of the FM LIDAR sensor system a high amount of the light that hit the object.

[0046] Regardless of the object's reflectivity, an FM LIDAR sensor system may be able to detect (e.g., classify, recognize, discover, etc.) the object at greater distances (e.g., 2×) than a conventional LIDAR sensor system. For example, an FM LIDAR sensor system may detect a low reflectively object beyond 300 meters, and a high reflectivity object beyond 400 meters.

[0047] To achieve such improvements in detection capability, the FM LIDAR sensor system may use sensors (e.g., sensors **130** in FIG. **1**). In some implementations, these sensors can be single photon sensitive, meaning that they can detect the smallest amount of light possible. While an FM LIDAR sensor system may, in some applications, use infrared wavelengths (e.g., 950 nm, 1550 nm, etc.), it is not limited to the infrared wavelength range (e.g., near infrared: 800 nm-1500 nm; middle infrared: 1500 nm-5600 nm; and far infrared: 5600 nm-1,000,000 nm). By operating the FM or PM LIDAR sensor system in infrared wavelengths, the FM or PM LIDAR sensor system can broadcast stronger light pulses or light beams than conventional LIDAR sensor systems.

[0048] Thus, by detecting an object at greater distances, an FM LIDAR sensor system may have more time to react to unexpected obstacles. Indeed, even a few milliseconds of extra time could improve response time and comfort, especially with heavy vehicles (e.g., commercial trucking vehicles) that are driving at highway speeds.

[0049] The FM LIDAR sensor system can provide accurate velocity for each data point instantaneously. In some implementations, a velocity measurement is accomplished using the Doppler effect which shifts frequency of the light received from the object based at least one of the velocity in the radial direction (e.g., the direction vector between the object detected and the sensor) or the frequency of the laser signal. For example, for velocities encountered in on-road situations where the velocity is less than 100 meters per second (m/s), this shift at a wavelength of 1550 nanometers (nm) amounts to the frequency shift that is less than 130 megahertz (MHz). This frequency shift is small such that it is difficult to detect directly in the optical domain. However, by using coherent detection in FMCW, PMCW, or FMQW LIDAR sensor systems, the signal can be converted to the RF domain such that the frequency shift can be calculated using various signal processing techniques. This enables the autonomous vehicle control system to process incoming data faster.

[0050] Instantaneous velocity calculation also makes it easier for the FM LIDAR sensor system to determine distant or sparse data points as objects and/or track how those objects are moving over time. For example, an FM LIDAR sensor (e.g., sensors **130** in FIG. **1**) may only receive a few

returns (e.g., hits) on an object that is 300 m away, but if those return give a velocity value of interest (e.g., moving towards the vehicle at >70 mph), then the FM LIDAR sensor system and/or the autonomous vehicle control system may determine respective weights to probabilities associated with the objects.

[0051] Faster identification and/or tracking of the FM LIDAR sensor system gives an autonomous vehicle control system more time to maneuver a vehicle. A better understanding of how fast objects are moving also allows the autonomous vehicle control system to plan a better reaction.

[0052] The FM LIDAR sensor system can have less static compared to conventional LIDAR sensor systems. That is, the conventional LIDAR sensor systems that are designed to be more light-sensitive typically perform poorly in bright sunlight. These systems also tend to suffer from crosstalk (e.g., when sensors get confused by each other's light pulses or light beams) and from self-interference (e.g., when a sensor gets confused by its own previous light pulse or light beam). To overcome these disadvantages, vehicles using the conventional LIDAR sensor systems often need extra hardware, complex software, and/or more computational power to manage this “noise.”

[0053] In contrast, FM LIDAR sensor systems do not suffer from these types of issues because each sensor is specially designed to respond only to its own light characteristics (e.g., light beams, light waves, light pulses). If the returning light does not match the timing, frequency, and/or wavelength of what was originally transmitted, then the FM sensor can filter (e.g., remove, ignore, etc.) out that data point. As such, FM LIDAR sensor systems produce (e.g., generates, derives, etc.) more accurate data with less hardware or software requirements, enabling smoother driving.

[0054] The FM LIDAR sensor system can be easier to scale than conventional LIDAR sensor systems. As more self-driving vehicles (e.g., cars, commercial trucks, etc.) show up on the road, those powered by an FM LIDAR sensor system likely will not have to contend with interference issues from sensor crosstalk. Furthermore, an FM LIDAR sensor system uses less optical peak power than conventional LIDAR sensors. As such, some or all of the optical components for an FM LIDAR can be produced on a single chip, which produces its own benefits, as discussed herein.

2.1 Commercial Trucking

[0055] FIG. 2 is a block diagram illustrating an example of a system environment for autonomous commercial trucking vehicles, according to some implementations. The environment **100B** includes a commercial truck **102B** for hauling cargo **106B**. In some implementations, the commercial truck **102B** may include vehicles configured to long-haul freight transport, regional freight transport, intermodal freight transport (i.e., in which a road-based vehicle is used as one of multiple modes of transportation to move freight), and/or any other road-based freight transport applications. In some implementations, the commercial truck **102B** may be a flatbed truck, a refrigerated truck (e.g., a reefer truck), a vented van (e.g., dry van), a moving truck, etc. In some implementations, the cargo **106B** may be goods and/or produce. In some implementations, the commercial truck **102B** may include a trailer to carry the cargo **106B**, such as a flatbed trailer, a lowboy trailer, a step deck trailer, an extendable flatbed trailer, a sidekit trailer, etc.

[0056] The environment **100B** includes an object **110B** (shown in FIG. 2 as another vehicle) that is within a distance range that is equal to or less than 30 meters from the truck.

[0057] The commercial truck **102B** may include a LIDAR sensor system **104B** (e.g., an FM LIDAR sensor system, vehicle control system **120** in FIG. 1, LIDAR sensor system **500** in FIG. 5) for determining a distance to the object **110B** and/or measuring the velocity of the object **110B**. Although FIG. 2 shows that one LIDAR sensor system **104B** is mounted on the front of the commercial truck **102B**, the number of LIDAR sensor system and the mounting area of the LIDAR sensor system on the commercial truck are not limited to a particular number or a particular area. The commercial truck **102B** may include any number of LIDAR sensor systems **104B** (or components thereof, such as sensors, modulators, coherent signal generators, etc.) that are mounted onto any area (e.g., front, back, side, top, bottom, underneath, and/or bottom) of the commercial truck **102B** to facilitate the detection of an object in any free-space relative to the commercial truck

102B.

[0058] As shown, the LIDAR sensor system **104B** in environment **100B** may be configured to detect an object (e.g., another vehicle, a bicycle, a tree, street signs, potholes, etc.) at short distances (e.g., 30 meters or less) from the commercial truck **102B**.

[0059] FIG. **3** is a block diagram illustrating an example of a system environment for autonomous commercial trucking vehicles, according to some implementations. The environment **100C** includes the same components (e.g., commercial truck **102B**, cargo **106B**, LIDAR sensor system **104B**, etc.) that are included in environment **100B**.

[0060] The environment **100C** includes an object **110C** (shown in FIG. **3** as another vehicle) that is within a distance range that is (i) more than 30 meters and (ii) equal to or less than 150 meters from the commercial truck **102B**. As shown, the LIDAR sensor system **104B** in environment **100C** may be configured to detect an object (e.g., another vehicle, a bicycle, a tree, street signs, potholes, etc.) at a distance (e.g., 100 meters) from the commercial truck **102B**.

[0061] FIG. **4** is a block diagram illustrating an example of a system environment for autonomous commercial trucking vehicles, according to some implementations. The environment **100D** includes the same components (e.g., commercial truck **102B**, cargo **106B**, LIDAR sensor system **104B**, etc.) that are included in environment **100B**.

[0062] The environment **100D** includes an object **110D** (shown in FIG. **4** as another vehicle) that is within a distance range that is more than 150 meters from the commercial truck **102B**. As shown, the LIDAR sensor system **104B** in environment **100D** may be configured to detect an object (e.g., another vehicle, a bicycle, a tree, street signs, potholes, etc.) at a distance (e.g., 300 meters) from the commercial truck **102B**.

[0063] In commercial trucking applications, it is important to effectively detect objects at all ranges due to the increased weight and, accordingly, longer stopping distance required for such vehicles. FM LIDAR sensor systems (e.g., FMCW and/or FMQW systems) or PM LIDAR sensor systems are well-suited for commercial trucking applications due to the advantages described above. As a result, commercial trucks equipped with such systems may have an enhanced ability to move both people and goods across short or long distances. In various implementations, such FM or PM LIDAR sensor systems can be used in semi-autonomous applications, in which the commercial truck has a driver and some functions of the commercial truck are autonomously operated using the FM or PM LIDAR sensor system, or fully autonomous applications, in which the commercial truck is operated entirely by the FM or LIDAR sensor system, alone or in combination with other vehicle systems.

3. LIDAR Sensor Systems

[0064] FIG. **5** depicts an example of a LIDAR sensor system **500**. The LIDAR sensor system **500** can be used to determine parameters regarding objects, such as range and velocity, and output the parameters to a remote system. For example, the LIDAR sensor system **500** can output the parameters for use by a vehicle controller that can control operation of a vehicle responsive to the received parameters (e.g., vehicle controller **598**) or a display that can present a representation of the parameters. The LIDAR sensor system **500** can be a coherent detection system. The LIDAR sensor system **500** can be used to implement various features and components of the systems described with reference to FIGS. **1-4**. The LIDAR sensor system **500** can include components for performing various detection approaches, such as to be operated as an amplitude modular LIDAR system or a coherent LIDAR system. The LIDAR sensor system **500** can be used to perform time of flight range determination. In some implementations, various components or combinations of components of the LIDAR sensor system **500**, such as laser source **504** and modulator **514**, can be in a same housing, provided in a same circuit board or other electronic component, or otherwise integrated. In some implementations, various components or combinations of components of the LIDAR sensor system **500** can be provided as separate components, such as by using optical couplings (e.g., optical fibers) for components that generate and/or receive optical signals, such as

light beams, or wired or wireless electronic connections for components that generate and/or receive electrical (e.g., data) signals.

[0065] The LIDAR sensor system **500** can include a laser source **504** that generates and emits a beam **506**, such as a carrier wave light beam. A splitter **508** can split the beam **506** into a beam **510** and a reference beam **512** (e.g., reference signal). In some implementations, any suitable optical, electronic, or opto-electronic elements can be used to provide the beam **510** and the reference beam **512** from the laser source **504** to other elements.

[0066] A modulator **514** can modulate one or more properties of the input beam **510** to generate a beam **516** (e.g., target beam). In some implementations, the modulator **514** can modulate a frequency of the input beam **510** (e.g., optical frequency corresponding to optical wavelength, where $c=\lambda v$, where c is the speed of light, λ is the wavelength, and v is the frequency). For example, the modulator **514** can modulate a frequency of the input beam **510** linearly such that a frequency of the beam **516** increases or decreases linearly over time. As another example, the modulator **514** can modulate a frequency of the input beam **510** non-linearly (e.g., exponentially). In some implementations, the modulator **514** can modulate a phase of the input beam **510** to generate the beam **516**. However, the modulation techniques are not limited to the frequency modulation and the phase modulation. Any suitable modulation techniques can be used to modulate one or more properties of a beam. Returning to FIG. 5, the modulator **514** can modulate the beam **510** subsequent to splitting of the beam **506** by the splitter **508**, such that the reference beam **512** is unmodulated, or the modulator **514** can modulate the beam **506** and provide a modulated beam to the splitter **508** for the splitter **508** to split into a target beam and a reference beam.

[0067] The beam **516**, which is used for outputting a transmitted signal, can have most of the energy of the beam **506** outputted by the laser source **504**, while the reference beam **512** can have significantly less energy, yet sufficient energy to enable mixing with a return beam **548** (e.g., returned light) scattered from an object. The reference beam **512** can be used as a local oscillator (LO) signal. The reference beam **512** passes through a reference path and can be provided to a mixer **560**. An amplifier **520** can amplify the beam **516** to output a beam **522**.

[0068] The LIDAR sensor system **500** can include an optic module **524**, which can receive the beam **522**. The optic module **524** can be a free space optic. For example, the optic module **524** can include one or more optics (e.g., lenses, mirrors, waveguides, grating couplers, prisms, waveplates) arranged to have a gap (e.g., air gap) between the one or more optics, allowing for free space transmission of light (e.g., rather than all light being coupled between optics by fibers). The optic module **524** can perform functions such as collimating, filtering, and/or polarizing the beam **522** to output a beam **530** to optics **532** (e.g., scanning optics).

[0069] Referring to FIG. 6, the optic module **524** can include at least one collimator **604** and at least one circulator **608**. For example, the circulator **608** can be between the collimator **604** and the optics **532** of FIG. 5. The circulator **608** can receive a collimated beam **612** outputted by the collimator **604** and output a beam **616** (e.g., the beam **530** depicted in FIG. 5) to the optics **532**. In some implementations, the circulator **608** can be between the laser source **504** and the collimator **604**. At least one of the collimator **604** or the circulator **608** can be free space optics (and can be coupled with one another in free space), such as by being optically coupled via air gaps rather than optical fibers.

[0070] Referring further to FIG. 5, the optic module **524** can receive return beam **548** from the optics **532** and provide the return beam **548** to the mixer **560**. The optics **532** can be scanning optics, such as one or more steering mirrors or polygon reflectors or deflectors to adjust the angle of received beams relative to outputted beams based on the orientation of outer surfaces (e.g., facets) of the optics relative to the received beam, or solid-state components (e.g., phased arrays, electro-optic crystals) configured to modify the direction of received light.

[0071] The optics **532** can define a field of view **544** that corresponds to angles scanned (e.g., swept) by the beam **542** (e.g., a transmitted beam). For example, the beam **542** can be scanned in

the particular plane, such as an azimuth plane or elevation plane (e.g., relative to an object to which the LIDAR sensor system **500** is coupled, such as an autonomous vehicle). The optics **532** can be oriented so that the field of view **544** sweeps an azimuthal plane relative to the optics **532**.

[0072] At least one motor **540** can be coupled with the optics **532** to control at least one of a position or an orientation of the optics **532** relative to the beam **530**. For example, where the optics **532** include a reflector or deflector, the motor **540** can rotate the optics **532** so that surfaces of the optics **532** at which the beam **530** is received vary in angle or orientation relative to the beam **530**, causing the beam **542** to be varied in angle or direction as the beam **542** is outputted from the optics **532**.

[0073] The beam **542** can be outputted from the optics **532** and reflected or otherwise scattered by an object (not shown) as a return beam **548** (e.g., return signal). The return beam **548** can be received on a reception path, which can include the circulator **528**, and provided to the mixer **560**.

[0074] The mixer **560** can be an optical hybrid, such as a 90 degree optical hybrid. The mixer **560** can receive the reference beam **512** and the return beam **548**, and mix the reference beam **512** and the return beam **548** to output a signal **564** responsive to the reference beam **512** and the return beam **548**. The signal **564** can include an in-phase (I) component **568** and a quadrature (Q) component **572**.

[0075] The LIDAR sensor system **500** can include a receiver **576** that receives the signal **564** from the mixer **560**. The receiver **576** can generate a signal **580** responsive to the signal **564**, which can be an electronic (e.g., radio frequency) signal. The receiver **576** can include one or more photodetectors that output the signal **580** responsive to the signal **564**.

[0076] The LIDAR sensor system **500** can include a processing system **590**, which can be implemented using features of the vehicle control system **120** described with reference to FIG. 1. The processing system **590** can process data received regarding the return beam **548**, such as the signal **580**, to determine parameters regarding the object such as range and velocity. The processing system **590** can include a scanner controller **592** that can provide scanning signals to control operation of the optics **532**, such as to control the motor **540** to cause the motor **540** to rotate the optics **532** to achieve a target scan pattern, such as a sawtooth scan pattern or step function scan pattern. The processing system **590** can include a Doppler compensator **594** that can determine the sign and size of a Doppler shift associated with processing the return beam **548** and a corrected range based thereon along with any other corrections. The processing system **590** can include a modulator controller **596** that can send one or more electrical signals to drive the modulator **514**.

[0077] The processing system **590** can include or be communicatively coupled with a vehicle controller **598** to control operation of a vehicle for which the LIDAR sensor system **500** is installed (e.g., to provide complete or semi-autonomous control of the vehicle). For example, the vehicle controller **598** can be implemented by at least one of the LIDAR sensor system **500** or control circuitry of the vehicle. The vehicle controller **598** can control operation of the vehicle responsive to at least one of a range to the object or a velocity of the object determined by the processing system **590**. For example, the vehicle controller **598** can transmit a control signal to at least one of a steering system or a braking system of the vehicle to control at least one of speed or direction of the vehicle.

3.1 LIDAR Sensor System Including Grating Couplers for Multiple Direction Reception

[0078] FIG. 7 depicts a block diagram of an example of a LIDAR sensor system **700**. The LIDAR sensor system **700** can incorporate features of the LIDAR sensor system **500** and optic module **524** described with reference to FIGS. 5 and 6, respectively. The LIDAR sensor system **700** can facilitate pitch-catch compensation, i.e. accounting for time delays or other offsets resulting from the round-trip path of the transmit beam outputted from the LIDAR sensor system **700**, reflected or otherwise scattered by object(s), and then returned as the return beam to the LIDAR sensor system **700** for detection and processing, which might otherwise affect characteristics of the LIDAR sensor system **700** such as signal-to-noise ratio.

[0079] The LIDAR sensor system **700** can include a chip **705** on which various components of the LIDAR sensor system **700**, including transmitter **710** and receiver **712**, can be provided. For example, the chip **705** can be a photonic integrated chip, such that various components of the LIDAR sensor system **700** for generating, modulating, and processing optical signals and performing photonic operations are implemented by the chip **705**. The chip **705** can be a semiconductor circuit chip. The chip **705** can be made from at least one III-V semiconductor material. For example, the chip **705** can be made from a silicon material or pure silicon. The chip **705** can be made from gallium nitride. The chip **705** can be made from aluminum nitride. The chip **705** can be made from gallium nitride and pure silicon.

[0080] The LIDAR sensor system **700** includes at least one transmitter **710**. The transmitter **710** can receive a beam (e.g., various beams described with reference to FIGS. **5** and **6**) and output transmit beam **735** with particular characteristics such as direction, polarization, or various combinations thereof.

[0081] The transmitter **710** can include at least one grating coupler **715** (e.g., a first grating coupler). The grating coupler **715** can be a structure having a plurality of spaced apart channels, such as parallel channels, which may have various shapes of the same or differing sizes. The grating coupler **715** can be a structure formed by etching on the chip **705**. The grating coupler **715** can be a structure formed by deposition of material on the chip **705**.

[0082] The grating coupler **715** can be configured to couple light off of the chip **705**, such as into free space away from the chip **705**. For example, the grating coupler **715** can couple light off of the chip **705** in a two-dimensional pattern, such as a two-dimensional polarization. As such, the grating coupler **715** can output a transmit beam **735**, such as based on a beam outputted by laser source **504**.

[0083] The grating coupler **715** can output the transmit beam **735** to have a transmit polarization **795**. For example, various components upstream or downstream of the grating coupler **715** (e.g., optic module **524**) can be used to control the polarization of the transmit beam **735**.

[0084] The LIDAR sensor system **700** can include at least one scanner **740**, e.g., a steering mirror. For example, referring briefly to FIG. **5**, the scanner **740** can be coupled with the motor **540**, so that the scanner **740** can be rotated relative to the direction along which the transmit beam **735** is directed towards the scanner **740**.

[0085] The scanner **740** can scan bi-directionally. For example, the scanner **740** can scan in a first direction **760** and a second direction **765** (e.g., relative to axis **534** described with reference to FIG. **5**). The scanner **740** can receive the transmit beam **735** from the transmitter **710** and direct the transmit beam **735** towards the environment around the LIDAR sensor system **700**. As depicted in FIG. **7**, an object **702** can be present in the environment. The scanner **740** can receive a return beam **755** from reflection or scattering of the transmit beam **735** by the object **702**. In the time it takes for the transmit beam **735** to reach the object **702** and the return beam **755** to return from the object **702** to the scanner **740**, the scanner **740** may have rotated by a particular angle, as discussed more below. The scanner **740** can provide the return beam **755** to the receiver **712**.

[0086] The receiver **712** can include a plurality of grating couplers. For example, the receiver **712** can include a grating coupler **725** and a grating coupler **730**. The grating couplers **725**, **730** can be provided or formed in a manner similar to or identical to the grating coupler **715**. The grating couplers **715**, **725**, **730** can be arranged in an array on the chip **705** (e.g., at least two parallel spaced-apart lines can extend through each of the grating couplers **715**, **725**, **730**). The grating couplers **715**, **725**, **730** can be arranged in a focal plane of at least one of the scanner **740** or one or more optical components between the grating couplers **715**, **725**, **730** and the scanner **740**.

[0087] The grating couplers **725**, **730** can be spaced from the grating coupler **715**. For example, the grating coupler **725** can be spaced from the grating coupler **715** by a first spacing (e.g., first distance). The first spacing can be associated with a first target range from the scanner **740** for detecting the object **702**. For example, the first target range can be within a range of distances from

the scanner **740** for which an expected signal to noise ratio of determining at least one of range to or velocity of the object **702** is greater than a threshold signal to noise ratio. The first spacing can be between about 6 micrometers (μm) and about 20 μm . For example, the grating coupler **725** can be spaced from the grating coupler **715** by about 12 μm .

[0088] The grating coupler **730** can be spaced from the grating coupler **715** by a second spacing (e.g., second distance). The second spacing can be associated with a second target range from the scanner **740**. The second target range can be greater than the first target range. The second spacing can be between about 6 μm and about 20 μm . For example, the grating coupler **730** can be spaced from the grating coupler **715** by about 12 μm s. In some implementations, various spacings can be between about 20 μm and about 200 μm , such as with the use of external optics to facilitate directing of light by the LIDAR sensor system **700**.

[0089] The grating couplers **725**, **730** can receive the return beam **755** provided by the scanner **740**. The grating couplers **725**, **730** can couple light, e.g., the return beam **755**, from free space onto the chip **705**. As previously mentioned, the scanner **740** can rotate in the time it takes for the transmit beam **735** to travel to the object **702** and be returned as the return beam **755** to the receiver **712**, which can result in an angular displacement of the return beam **755** in the direction the scanner **740** is scanning. The angular displacement can manifest as a translation in the focal plane when the return beam **755** is provided to the receiver **712**. The translation can be referred to as focal plane drift. Since the grating couplers **725**, **730** can couple light on the chip **705** and are distinct and spatially separated, the grating couplers **725**, **730** can be provided for two directions of focal plane drift. Additionally, the translational distance can be optimized for a target time for the transmit beam **735** to travel to the object **702** and the return beam **755** to return from the object **702** to the receiver **712**, and thus a specific range to the object **702** (because the velocity of the transmit beam **735** and the return beam **755** is known).

[0090] The return beam **755** can have components associated with various polarizations, such as based on how the transmit beam **735** is outputted and/or passed through devices such as optic module **524**. As such, the scanner **740** can provide a first component **770** of the return beam **755**, which can be associated with a first polarization **780**, and the grating coupler **725** can receive the first component **770** of the return beam **755**. The scanner **740** can provide a second component **775** of the return beam **755** associated with a second polarization **785**, and the grating coupler **730** can receive the second component **775**.

[0091] The first polarization **780** can be different than the second polarization **785**. The transmit polarization **795** can be the same as the first polarization **780**. For example, the grating coupler **715** can output the transmit beam **735** at the first polarization **780**. The grating couplers **725**, **730** can receive the return beam **755** at the first polarization **780** or at the second polarization **785**. The second polarization **785** can be orthogonal to the first polarization **780**.

[0092] The grating couplers **725**, **730** can be configured to only receive a single polarization of a beam, e.g., light. The grating couplers **725**, **730** can receive light of different polarizations. For example, the grating coupler **725** can receive light of the same polarization as that of the transmit beam **735** and the grating coupler **730** can receive light of a polarization orthogonal to the polarization of the transmit beam **735**. The grating couplers **725**, **730** can be configured to receive the same polarization of light as each other. For example, the grating couplers **725**, **730** can receive light (only) of a polarization orthogonal to the polarization of the transmit beam **735**, as depicted in FIG. 7, or that is the same as the polarization of the transmit beam **735**.

[0093] The receiver **712** can output at least one signal based on the first component **770** of the return beam **755** received by the grating coupler **725** and based on the second component **775** of the return beam **755** received by the grating coupler **730**. The at least one signal outputted by the receiver **712** can be used by various systems described herein, such as the vehicle control system **120**, to determine at least one of a range to or a velocity of the object **702**, such as to control operation of an autonomous vehicle responsive to the at least one of the range or the velocity.

[0094] FIG. 8 is a block diagram of an example of optical components of a system **800**. The system **800** can include components of and/or be used to implement various LIDAR sensor systems described herein, such as the transmitter **710** and the receiver **712**. For example, the system **800** can be used to implement a receiver of a single polarization, and a switched local oscillator.

[0095] As depicted in FIG. 8, the system **800** can include a transmit antenna **810**, which can be implemented by the grating coupler **715** of the transmitter **710**. For example, the transmit antenna **810** can be an optical antenna integrated onto the chip **705**. The transmit antenna **810** can couple the transmit beam **735** into free space. The transmit antenna **810** can be oriented at an angle relative to a surface on which the transmit antenna **810** is provided corresponding to the polarization of the transmit beam **735**.

[0096] The system **800** can include a local oscillator **815**. The local oscillator **815** can output local oscillator (LO) signals. For example, the local oscillator **815** can actively switch the LO signal between at least two mixers, as discussed more below, based on the scanning direction of the scanner **740**. For example, the local oscillator **815** can output a first LO signal and a second LO signal. The LO signals can be similar to or the same as reference beam **512**. For example, the local oscillator **815** can transmit the first LO signal, e.g., a reference beam **820**. The local oscillator **815** can transmit the second LO signal, e.g., a reference beam **830**. By the local oscillator **815** switching the LO signal, the amount of power required by the local oscillator **815** can be reduced.

[0097] The system **800** can include a receive antenna **825** of the receiver **712**. The receive antenna **825** can be implemented by the grating coupler **725** of the receiver **712**. For example, the receive antenna **825** can be an optical antenna integrated onto the chip **705**. The receive antenna **825** can couple the first component **770** of the return beam **755** onto the chip **705** from free space. The receive antenna **825** can be oriented at an angle relative to a surface on which the receive antenna **825** is provided corresponding to the first polarization **780** of the first component **770** of the return beam **755**.

[0098] The receive antenna **825** can receive the return beam **755**. For example, the receive antenna **825** can receive the first component **770** of the return beam **755**. As such, the receive antenna **825** can be oriented at the same polarization as the first component **770** of the return beam **755**. For example, the receive antenna **825** can have the same polarization as the first polarization **780**. The receive antenna **825** can provide the first component **770** of the return beam **755** to other elements of the system **800**, as discussed more below.

[0099] The system **800** can include a receive antenna **835** of the receiver **712**. The receive antenna **835** can be implemented by the grating coupler **730** of the receiver **712**. For example, the receive antenna **835** can be an optical antenna integrated onto the chip **705**. The receive antenna **835** can couple the second component **775** of the return beam **755** onto the chip **705** from free space. The receive antenna **835** can be oriented at an angle relative to a surface on which the receive antenna **835** is provided corresponding to the second polarization **785** of the second component **775** of the return beam **755**.

[0100] The receive antenna **835** can receive the return beam **755**. For example, the receive antenna **835** can receive the second component **775** of the return beam **755**. As such, the receive antenna **835** can be oriented at the same polarization as the second component **775** of the return beam **755**. For example, the receive antenna **835** can have the same polarization as the second polarization **785**. The receive antenna **835** can provide the second component **775** of the return beam **755** to other elements of the system **800**, as discussed more below.

[0101] The system **800** can include at least one mixer. The mixer can be similar to or the same as the mixer **560**. For example, the system **800** can include a mixer **840**, which functions similar to the mixer **560**. For example, the system **800** can include a mixer **845**, which functions similar to the mixer **560**. The mixers **840**, **845** can each be a 2×2 optical mixer. The mixers **840**, **845** can each be an optical hybrid. For example, the mixers **840**, **845** can each be a 90 degree optical hybrid. For example, the mixers **840**, **845** can each be a 2×4 optical hybrid.

[0102] The mixers **840**, **845** can receive signals. For example, the mixer **840** can receive the reference beam **820**, e.g., the first LO signal. For example, the mixer **840** can receive the first component **770** of the return beam **755** from the receive antenna **825**. For example, the mixer **845** can receive the reference beam **830**, e.g., the second LO signal. For example, the mixer **845** can receive the second component **775** of the return beam **755** from the receive antenna **835**.

[0103] The mixers **840**, **845** can each output a signal. For example, the output signals can be based on the signals that each of the mixers **840**, **845** received from the receive antennas **825**, **835**. For example, the output signals can be based on the signals that each of the mixers **840**, **845** received from the local oscillator **815**. For example, the mixers **840**, **845** can output signals responsive to the return beam **755** and the reference beams **820**, **830**. For example, the mixers **840**, **845** can mix the return beam **755** and the reference beams **820**, **830** and each output a signal. The mixer **840** can output a signal **850**. The signal **850** can be responsive to and based on the first component **770** of the return beam **755** and the reference beam **820**. The mixer **845** can output a signal **855**. The signal **855** can be responsive to and based on the second component **775** of the return beam **755** and the reference beam **830**. The mixers **840**, **845** can provide the signals **850**, **855** to the grating couplers **725**, **730**, respectively, such as for the grating couplers **725**, **730** to provide the components **770**, **775** of the return beam **755** to optical detection devices, such as one or more photodetectors of the receiver **712**.

[0104] FIG. **9** is a block diagram of an example of a LIDAR sensor system **900** including the optical components of FIG. **8**. The LIDAR sensor system **900** can incorporate features of the LIDAR sensor system **500** and optic module **524** described with reference to FIGS. **5** and **6**, respectively. The LIDAR sensor system **900** can facilitate pitch-catch compensation, including accounting for time delays or other offsets resulting from the round-trip path of the transmit beam outputted from the LIDAR sensor system **900**, reflected or otherwise scattered by object(s), and then returned as the return beam to the LIDAR sensor system **900** for detection and processing, which might otherwise affect characteristics of the LIDAR sensor system **900** such as signal-to-noise ratio.

[0105] The LIDAR sensor system **900** can include the grating coupler **715** configured to output the transmit beam **735**. The LIDAR sensor system **900** can include the scanner **740** configured to receive the transmit beam **735** from the transmitter **710**, provide the transmit beam **735** to the environment, and receive the return beam **755** of reflection of the transmit beam **735** from the object **702**. The LIDAR sensor system **900** can include the grating coupler **725** and the grating coupler **730**. The grating coupler **725** can receive the first component **770** of the return beam **755** at the first polarization **780**. The grating coupler **730** can receive the second component **775** of the return beam **755** at the second polarization **785**.

[0106] The first polarization **780** can be the same as the second polarization **785**. The transmit polarization **795** can be different than the first polarization **780** and the second polarization **785**. For example, the grating coupler **715** can output the transmit beam **735** at the transmit polarization **795** and the grating couplers **725**, **730** can receive the components **770**, **775** of the return beam **755** with the polarizations **780**, **785** each orthogonal to the transmit polarization **795**.

[0107] The LIDAR sensor system **900** can include a displacer **905**. The transmit beam **735** can pass unaffected through the displacer **905**. The displacer **905** can be a birefringent displacer such that the displacer **905** has two different refractive indices (various features and examples of the displacer **905** are described further herein). The displacer **905** can displace the return beam **755**. For example, the displacer **905** can displace the return beam **755** in the opposite polarization to the transmit beam **735** by a fixed amount. For example, the displacer **905** can displace the return beam **755** to the right, e.g., relative to the orientation of the array during operation, of the transmit antenna **810** upon receive. The receive antennas **825**, **835** are aligned with the polarizations **780**, **785** of the components **770**, **775** of the return beam **755**. The receive antennas **825**, **835** are located on either side of the displaced return beam **755** such that they compensate for the focal plane drift,

discussed above, at the specific target distance, e.g., the range of the object **702** to the scanner **740**.
[0108] The LIDAR sensor system **900** can include optic module **524** (e.g., collimator **604** of optic module **524**). The collimator **604** can be positioned between the transmitter **710** and the scanner **740**. The transmit beam **735** can pass unaffected through the collimator **604**. For example, the collimator **604** can be configured to provide the transmit beam **735** to the scanner **740**. The collimator **604** can be configured to collimate the transmit beam **735**. The collimator **604** can be configured to provide the collimated beam **910** to other components of the LIDAR sensor system **900**, as discussed more below. For example, the collimator **604** can be configured to provide the collimated beam **910** to the scanner **740**.

[0109] The LIDAR sensor system **900** can include a wave plate **915**. The wave plate **915** can be a quarter wave plate. The transmit beam **735** can pass unaffected through the wave plate **915**. The wave plate **915** can induce circular polarization. For example, the wave plate **915** can circularly polarize the components **770**, **775** of the return beam **755**.

[0110] FIG. **10** is a block diagram of an example of optical components of a system **1000**. The system **1000** can include components of and/or be used to implement various LIDAR sensor systems described herein, such as the transmitter **710** and the receiver **712**. For example, the system **1000** can be used to implement a receiver of a single polarization, and a switched local oscillator.

[0111] The system **1000** is similar to the system **800**. However, the polarizations of the receive antennas **825**, **835** are aligned with the polarization of the transmit antenna **810**. As mentioned above, the transmit antenna **810** can have the same polarization as the transmit polarization **795**, the receive antenna **825** can have the same polarization as the first polarization **780**, and the receive antenna **835** can have the same polarization as the second polarization **785**. In system **1000**, the transmit polarization **795** can be the same as the polarizations **780**, **785** such that the transmit antenna **810** can have the same polarization as the receive antennas **825**, **835**.

[0112] These receive antennas **825**, **835** can be positioned on either side of the transmit antenna **810**. For example, the receive antennas **825**, **835** and the transmit antenna **810** can be positioned along the direction of the focal plane drift, discussed above. The physical proximity of the receive antennas **825**, **835** to the transmit antenna **810** is selected to compensate for the focal plane drift for the specific target distance, e.g., the range of the object **702** to the scanner **740**.

[0113] FIG. **11** is a block diagram of an example of a LIDAR sensor system **1100** including the optical components of FIG. **10**. The LIDAR sensor system **1100** can incorporate features of the LIDAR sensor system **500** and optic module **524** described with reference to FIGS. **5** and **6**, respectively. The LIDAR sensor system **1100** can facilitate pitch-catch compensation, i.e. accounting for time delays or other offsets resulting from the round-trip path of the transmit beam outputted from the LIDAR sensor system **1100**, reflected or otherwise scattered by object(s), and then returned as the return beam to the LIDAR sensor system **1100** for detection and processing, which might otherwise affect characteristics of the LIDAR sensor system **1100** such as signal-to-noise ratio.

[0114] Because the system **1000** is similar to the system **800**, except for the polarizations of the receive antennas **825**, **835** being aligned with the polarization of the transmit antenna **810**, the block diagram depicted in FIG. **11** is similar to the block diagram depicted in FIG. **9**. However, since the polarizations of the receive antennas **825**, **835** are not opposite of the polarization to the transmit antenna **810**, the displacer **905** can be omitted. For example, in the LIDAR sensor system **900**, the displacer **905** can displace the return beam **755** in the opposite polarization to the transmit beam **735** by a fixed amount. However, in the LIDAR sensor system **1100**, the displacement of the return beam **755** it is not needed when the polarizations **780**, **785**, **795** are the same.

[0115] FIG. **12** is a block diagram of an example of optical components of a system **1200**. The system **1200** can include components of and/or be used to implement various LIDAR sensor systems described herein, such as the transmitter **710** and the receiver **712**. For example, the system **1200** can be used to implement a receiver of a single polarization, and a switched local oscillator.

[0116] The system **1200** is similar to the system **800**. However, the mixers **840**, **845** are in communication with two independent and balanced photodetectors, instead of the single photodetector, e.g., the receiver **712**, with two photodiodes, e.g., the grating couplers **725**, **730**. As such, the system **1200** can include a receiver **1205**, e.g., a second receiver. The receiver **1205** can include the grating coupler **730**. The receiver **712** can include the grating coupler **725**. The mixer **840** can provide the signal **850** to the receiver **712**, similar to the system **800**. However, in the system **1200**, the mixer **845** can provide the signal **855** to the receiver **1205**.

[0117] In the system **1200**, the local oscillator **815** can be passively split between the mixer **840** and the mixer **845**. For example, the local oscillator **815** can actively switch the LO signal between the mixer **840** and the mixer **845** based on the scanning direction of the scanner **740**. For example, the local oscillator **815** can transmit the reference beam **820**, e.g., the first LO signal, to the mixer **840**. For example, the local oscillator **815** can transmit the reference beam **830**, e.g., the second LO signal, to the mixer **845**.

[0118] FIG. **13** is a block diagram of an example of optical components of a system **1300**. The system **1300** can include components of and/or be used to implement various LIDAR sensor systems described herein, such as the transmitter **710** and the receiver **712**. For example, the system **1300** can be used to implement a receiver of a single polarization, and a switched local oscillator.

[0119] The system **1300** is similar to the system **1000** such that the displacer **905** can be omitted during operation. However, similar to the system **1200**, the mixers **840**, **845** are in communication with two independent and balanced photodetectors, instead of the single photodetector, e.g., the receiver **712**, with two photodiodes, e.g., the grating couplers **725**, **730**. As such, the system **1300** can include the receiver **1205**. The receiver **1205** can include the grating coupler **730**. The receiver **712** can include the grating coupler **725**. The mixer **840** can provide the signal **850** to the receiver **712**, similar to the system **800**. However, in the system **1300**, the mixer **845** can provide the signal **855** to the receiver **1205**. Further, similarly to the system **1200**, the local oscillator **815** can be passively split between the mixer **840** and the mixer **845**.

[0120] FIG. **14** is a block diagram of an example of optical components of a system **1400**. The system **1400** can include components of and/or be used to implement various LIDAR sensor systems described herein, such as the transmitter **710** and the receiver **712**. For example, the system **1400** can be used to implement a receiver of a single polarization, and a switched local oscillator.

[0121] The system **1400** is a combination of the system **800** and the system **1000**. As such, the system **1400** can include two separate local oscillators. For example, the system **1400** can include the local oscillator **815** and a local oscillator **1405**, e.g., a second local oscillator. The local oscillator **1405** can function similar to or the same as the local oscillator **815**. For example, the local oscillator **1405** can output LO signals. For example, the local oscillator **1405** can actively switch the LO signal between at least to mixers based on the scanning direction of the scanner **740**. For example, the local oscillator **1405** can output a third LO signal and a fourth LO signal. The LO signals can be similar to or the same as reference beam **512**. For example, the local oscillator **1405** can transmit the third LO signal, e.g., a reference beam **1410**. For example, the local oscillator **1405** can transmit the fourth LO signal, e.g., a reference beam **1420**. By the local oscillator **1405** switching the LO signal, the amount of power required by the local oscillator **1405** can be reduced.

[0122] The system **1400** can include a receive antenna **1415** and a receive antenna **1425**. The receive antennas **1415**, **1425** are similar to the receive antennas **825**, **835**. For example, the receive antennas **1415**, **1425** can receive the return beam **755**, as discussed more below.

[0123] The system **1400** can include a mixer **1430** and a mixer **1435**. The mixers **1430**, **1435** can be similar to or the same as the mixers **840**, **845**. The mixers **1430**, **1435** can receive signals, as discussed more below. The mixer **1430** can output a signal **1440**, as discussed more below. The mixer **1435** can output a signal **1445**, as discussed more below.

[0124] FIG. **15** is a block diagram of an example of a LIDAR sensor system **1500** including the optical components of FIG. **14**. The LIDAR sensor system **1500** can incorporate features of the

LIDAR sensor system **500** and optic module **524** described with reference to FIGS. **5** and **6**, respectively. The LIDAR sensor system **1500** can facilitate pitch-catch compensation, i.e. accounting for time delays or other offsets resulting from the round-trip path of the transmit beam outputted from the LIDAR sensor system **1500**, reflected or otherwise scattered by object(s), and then returned as the return beam to the LIDAR sensor system **1500** for detection and processing, which might otherwise affect characteristics of the LIDAR sensor system **1500** such as signal-to-noise ratio.

[0125] The LIDAR sensor system **1500** can include the displacer **905** and the collimator **604**, similar to the system **800** depicted in FIG. **8**. However, in the system **1400**, the LIDAR sensor system **1500** can include the wave plate **915**. The wave plate **915** can be a quarter wave plate. The transmit beam **735** can pass unaffected through the wave plate **915**. The wave plate **915** can induce circular polarization. For example, the wave plate **915** can circularly polarize components of the return beam **755**.

[0126] The LIDAR sensor system **1500** can include a grating coupler **1510** and a grating coupler **1520**. The grating couplers **1510**, **1520** can receive the return beam **755** provided by the scanner **740**. For example, the grating coupler **1510** can receive a third component **1505** of the return beam **755** provided by the scanner **740**. For example, the grating coupler **1520** can receive a fourth component **1525** of the return beam **755** provided by the scanner **740**. The third component **1505** of the return beam **755** can be at a third polarization **1515**. The fourth component **1525** of the return beam **755** can be at a fourth polarization **1530**.

[0127] As discussed above, the LIDAR sensor system **1500** can include the receive antenna **1415** and the receive antenna **1425** of the receiver **1205**. The receive antenna **1415** can be implemented by the grating coupler **1510** of the receiver **712**. For example, the receive antenna **1415** can be an optical antenna integrated onto the chip **705**. The receive antenna **1415** can couple the third component **1505** of the return beam **755** onto the chip **705** from free space. The receive antenna **1415** can be oriented at an angle relative to a surface on which the receive antenna **1415** is provided corresponding to the third polarization **1515** of the third component **1505** of the return beam **755**. The receive antenna **1425** can be implemented by the grating coupler **1520** of the receiver **712**. For example, the receive antenna **1425** can be an optical antenna integrated onto the chip **705**. The receive antenna **1425** can couple the fourth component **1525** of the return beam **755** onto the chip **705** from free space. The receive antenna **1425** can be oriented at an angle relative to a surface on which the receive antenna **1425** is provided corresponding to the fourth polarization **1530** of the fourth component **1525** of the return beam **755**.

[0128] The receive antenna **1415** can receive the return beam **755**. For example, the receive antenna **1415** can receive the third component **1505** of the return beam **755**. As such, the receive antenna **1415** can be oriented at the same polarization as the third component **1505** of the return beam **755**. For example, the receive antenna **1415** can have the same polarization as the third polarization **1515**. The receive antenna **1415** can provide the third component **1505** of the return beam **755** to the mixer **1430**.

[0129] The receive antenna **1425** can receive the return beam **755**. For example, the receive antenna **1425** can receive the fourth component **1525** of the return beam **755**. As such, the receive antenna **1425** can be oriented at the same polarization as the fourth component **1525** of the return beam **755**. For example, the receive antenna **1425** can have the same polarization as the fourth polarization **1530**. The receive antenna **1425** can provide the fourth component **1525** of the return beam **755** to the mixer **1435**.

[0130] As discussed above, the mixers **1430**, **1435** can receive signals. For example, the mixer **1430** can receive the reference beam **1410**, e.g., the third LO signal. For example, the mixer **1430** can receive the third component **1505** of the return beam **755** from the receive antenna **1415**. For example, the mixer **1435** can receive the reference beam **1420**, e.g., the fourth LO signal. For example, the mixer **1435** can receive the fourth component **1525** of the return beam **755** from the

receive antenna **1425**.

[0131] As discussed above, the mixers **1430**, **1435** can output signals. For example, the output signals can be based on the signals that each of the mixers **1430**, **1435** received from the receive antennas **1415**, **1425**, respectively. For example, the output signals can be based on the signals that each of the mixers **1430**, **1435** received from the local oscillator **1405**. For example, the mixers **1430**, **1435** can output signals responsive to the return beam **755** and the reference beams **1410**, **1420**. For example, the mixers **1430**, **1435** can mix the return beam **755** and the reference beams **1410**, **1420** and each output a signal.

[0132] The mixer **1430** can output the signal **1440**. The signal **1440** can be responsive to and based on the third component **1505** of the return beam **755** and the reference beam **1410**. The mixer **1435** can output the signal **1445**. The signal **1445** can be responsive to and based on the fourth component **1525** of the return beam **755** and the reference beam **1420**. The mixers **1430**, **1435** can provide the signals **1440**, **1445** to two photodiodes, e.g., the grating couplers **1510**, **1520**, respectively. The two photodiodes can have two physically separate optical inputs such that the mixers **1430**, **1435** can provide the signals **1440**, **1445** to a single photodetector. For example, the mixers **1430**, **1435** can provide the signals **1440**, **1445** to the receiver **1205**.

[0133] FIG. **16** is a block diagram of an example of optical components of a system **1600**. The system **1600** can include components of and/or be used to implement various LIDAR sensor systems described herein, such as the transmitter **710** and the receiver **712**. For example, the system **1600** can be used to implement a receiver of a single polarization, and a switched local oscillator.

[0134] The system **1600** is similar to the system **1400**. However, similarly to the system **1200**, the mixers **1430**, **1435** are in communication with two independent and balanced photodetectors, instead of the single photodetector, e.g., the receiver **1205**. As such, the system **1600** can include a receiver **1605**, e.g., a third receiver, and a receiver **1610** e.g., a fourth receiver. The receiver **1605** can include the grating coupler **1510**. The receiver **1610** can include the grating coupler **1520**. Additionally, the receiver **712** can include the grating coupler **725** and the receiver **1205** can include the grating coupler **730**. The mixer **840** can provide the signal **850** to the receiver **712**. The mixer **845** can provide the signal **855** to the receiver **1205**. The mixer **1430** can provide the signal **1440** to the receiver **1605**. The mixer **1435** can provide the signal **1445** to the receiver **1610**.

3.2 LIDAR Sensor System Including Circulator

[0135] As discussed above, the LIDAR sensor system **500** can use various optical components to generate, encode information onto, and output light to be reflected or otherwise scattered by objects in the environment around the LIDAR sensor system **500**. This can include, for example, at least some transmission of the light through free space between components of the LIDAR sensor system **500**, including in implementations in which the light is provided from a laser source and modulated in frequency and/or phase using on-chip components, and then directed off of the chip by collimation and/or circulator optics to scanning optics to be directed into the environment. Systems and methods in accordance with the present disclosure can implement various such components to perform free-space optical transmission, maintain or improve signal-to-noise ratio, and reduce the overall size of the circulator.

[0136] FIG. **17** depicts a block diagram of a circulator **1700**, which can be implemented in various LIDAR sensor systems described herein, such as the LIDAR sensor system **500** and the optics module **524**. The circulator **1700** can be implemented by the chip **705**. The chip **705** can include the transmitter **710** and the grating coupler **715**. The transmitter **710** can output the transmit beam **735**. The transmit beam **735** can be associated with a polarization. For example, the transmit beam **735** can be associated with the transmit polarization **795**. The transmitter **710** can be positioned on the chip **705**. For example, the transmitter **710** can be on the chip **705** and can output the transmit beam **735** out of a plane of the chip **705**.

[0137] The chip **705** can include the receiver **712**. The receiver **712** can include the grating coupler **725**. The grating coupler **725** can be spaced by a spacing **1720** from the transmitter **710**. The

spacing **1720** between the grating coupler **725** and the transmitter **710** can be between about 6 μm and about 20 μm . The spacing **1720** between the grating coupler **725** and the transmitter **710** can be between about 12 μm and about 16 μm . For example, the spacing **1720** can be 14 μm . The spacing **1720** can correspond to the time delay associated with the first target range for detecting the object **702**, as previously discussed. For example, the grating coupler **725** is positioned such that the grating coupler **725** aligns with the translation due to the time delay of the first component **770** of the return beam **755**. The receiver **712** can include the grating coupler **730**. The grating coupler **730** can be spaced by a spacing **1725** from the transmitter **710**. The spacing **1725** between the grating coupler **730** and the transmitter **710** can be between about 30 μm and about 50 μm . The spacing **1725** between the grating coupler **730** and the transmitter **710** can be between about 32 μm and about 36 μm . For example, the spacing **1725** can be 34 μm .

[0138] The circulator **1700** can include the optics **532**, e.g., one or more scanning optics. The optics **532** can have a scan axis (e.g., axis **534** described with reference to FIG. 5) about which the optics **532** rotate. The grating coupler **725** and the grating coupler **730** can be along the scan axis.

[0139] The circulator **1700** can be similar to or the same as the circulator **528**. The circulator **1700** includes a wave plate **1730**. The wave plate **1730** can be made from a birefringent material, e.g., quartz or a plastic, for which the index of refraction can be different for various polarizations of light along at least one particular axis through the material. The wave plate **1730** can be positioned between the transmitter **710** and the optics **532**. The wave plate **1730** can be a half-wave plate. For example, with the wave plate **1730** being a half-wave plate, the wave plate **1730** can shift or rotate the polarization direction of linearly polarized light. As compared with operation of the wave plate **915**, the wave plate **1730** can rotate the transmit polarization **795**. For example, the wave plate **1730** can rotate the transmit polarization **795** by 45 degrees.

[0140] The circulator **1700** includes the displacer **905**. The displacer **905** can be positioned between the wave plate **1730** and the optics **532**. The displacer **905** can be made of various materials, such as birefringent materials including but not limited to YVO.sub.4, optically configured to control the path of light through the displacer **905** based on polarization of the light. The thickness of the displacer **905** can be selected based on target amounts of displacement and the type of material used for the displacer **905**.

[0141] For example, the displacer **905** can be made of LiNbO.sub.3. The displacer **905** can have a thickness between about 0.5 millimeters and 0.7 millimeters. For example, the displacer **905** can have a thickness of 0.6 millimeters. The displacer **905** can displace the return beam **755**. For example, as the return beam **755** provided by the object **702** contacts the displacer **905**, one of the two different refractive indices of the displacer **905** can displace a portion, e.g., the second component **775**, of the return beam **755**.

[0142] The displacer **905** can displace the second component **775** of the return beam **755**. For example, the displacer **905** can displace the second component **775** of the return beam **755** such that the second component **775** is between about 32 μm and about 36 μm from the rotated transmit beam **1715**. For example, the displacer **905** can displace the second component **775** of the return beam **755** such that the second component **775** is 34 μm from the rotated transmit beam **1715**. For example, the displacer **905** can displace the second component **775** of the return beam **755** by between about 18 μm and about 22 μm . For example, the displacer **905** can displace the second component **775** of the return beam **755** by 20 μm . For example, the displacer **905** can displace the second component **775** of the return beam **755** by 20 μm such that the second component **775** is 20 μm from the first component **770** of the return beam **755**. Thus, since the second component **775** of the return beam **755** is displaced by about 18 μm and 22 μm , the spacing **1725** between the grating coupler **730** and the transmitter **710** can be such that the grating coupler **730** can be positioned to receive the second component **775** of the return beam **755**. For example, the spacing **1725** between the grating coupler **730** and the transmitter **710** corresponds to the time delay and the displacement of the second component **775** relative to the first component **770** by the circulator **1700**.

[0143] The circulator **1700** can include a wave plate **1735**. The wave plate **1735** can be positioned between the displacer **905** and the optics **532**. The wave plate **1735** can be used to address variations in the polarization of the rotated transmit beam **1715** from a target polarization.

[0144] The circulator **1700** can direct the transmit beam **735** to the optics **532**. For example, the circulator **1700** can direct the transmit beam **735** to the optics **532** in a direction parallel with a plane of the chip **705**. For example, the optics **532** can be tilted relative to the plane of the chip **705**, e.g., at an angle greater than about 8 degrees, to reduce back reflections with respect to light passing in free space around the circulator **1700**; the grating coupler **715** can be angled, e.g., by at least about 12 degrees, such that the optics **532** can be positioned parallel to the chip **705**.

Positioning the optics **532** parallel to the chip **705**, e.g., the plane of the chip **706**, can simplify mounting. For example, the parallel mechanical mounting of the optics **532** relative to chip **705** can facilitate manufacturing, such as by allowing the optics **532** and/or circulator **1700** to be coupled (e.g., bonded) directly to the chip **705**.

[0145] The circulator **1700** can receive the transmit beam **735** and provide the transmit beam **735** to the optics **532**. For example, the transmit beam **735** can pass through the wave plate **1730** and the displacer **905** and contact the optics **532**. The circulator **1700** can receive the return beam **755** from reflection of the transmit beam **735** by the object **702**. For example, the return beam **755** can contact the displacer **905**. The circulator **1700** can split the return beam **755** into at least the first component **770** and the second component **775**. For example, the circulator **1700** can split the return beam **755** into at least the first component **770** and the second component **775** via the displacer **905**, as discussed above. The circulator **1700** can provide the first component **770** to the grating coupler **725** and the second component **775** to the grating coupler **730**. For example, the first component **770** and the second component **775** can pass through the wave plate **1730** to the grating coupler **725** and the grating coupler **730**, respectively.

[0146] The circulator **1700** can rotate the transmit beam **735** and provide the rotated transmit beam **1715** to the optics **532**. For example, the transmit beam **735** can contact the wave plate **1730** and the wave plate **1730** can rotate the transmit beam **735**. The wave plate **1730** can provide the rotated transmit beam **1715** to the displacer **905**. The rotated transmit beam **1715** can pass through the displacer **905** unaffected and contact the optics **532**.

[0147] The circulator **1700** can polarize the first component **770** of the return beam **755**. For example, the return beam **755** can return from the object **702** unpolarized. The return beam **755** can contact the displacer **905**. The displacer **905** can rotate, e.g., polarize, the first component **770** of the return beam **755** such that the polarization associated with the first component **770** corresponds with the polarization of the rotated transmit beam **1715**. The first component **770** of the return beam **755** can contact the wave plate **1730**. The wave plate **1730** can rotate, e.g., polarize, the first component **770** of the return beam **755** to provide the first component **770** at the first polarization **780**. The wave plate **1730** can rotate, e.g., polarize, the first component **770** of the return beam **755** such that the first polarization **780** associated with the first component **770** corresponds with the transmit polarization **795** associated with the transmit beam **735**. For example, the wave plate **1730** can rotate the first component **770** of the return beam **755** by 45 degrees.

[0148] The circulator **1700** can polarize the second component **775** of the return beam **755**. For example, the return beam **755** can return from the object **702** unpolarized. The return beam **755** can contact the displacer **905**. The displacer **905** can displace the second component **775** of the return beam **755**, as discussed above. The displacer **905** can rotate, e.g., polarize, the second component **775** of the return beam **755** such that the polarization associated with the second component **775** is orthogonal to the polarization of the rotated transmit beam **1715**. The second component **775** of the return beam **755** can contact the wave plate **1730**. The wave plate **1730** can rotate, e.g., polarize, the second component **775** of the return beam **755** to provide the second component **775** at the second polarization **785**. The wave plate **1730** can rotate, e.g., polarize, the second component **775** of the return beam **755** such that the second polarization **785** associated with the second component

775 is orthogonal to the transmit polarization 795 associated with the transmit beam 735. For example, the wave plate 1730 can rotate the second component 775 of the return beam 755 by 45 degrees. Since the polarization of the second component 775 is orthogonal to the first component 770 before the second component 775 and the first component 770 contact the wave plate 1730, the wave plate 1730 can rotate the second component 775 and the first component 770 each by 45 degrees in the same direction with the first polarization 780 and the second polarization 785 orthogonal to each other.

[0149] The circulator 1700 can provide the first component 770 of the return beam 755 with the first polarization 780 to the grating coupler 725. The circulator 1700 can provide the second component 775 of the return beam 755 with the second polarization 785 to the grating coupler 730. As discussed above, the grating coupler 725 and the grating coupler 730 can be rotated to correspond to the first polarization 780 and the second polarization 785, respectively, such as to be effectively aligned with the components 770, 775. The grating coupler 725 can receive at least a first threshold amount (e.g., greater than about thirty percent; between about thirty percent and about ninety percent; between about fifty percent and about eighty percent) of the first component 770 of the return beam 755. For example, the first threshold amount can correspond to an amount required by the grating coupler 725 to output a signal, e.g., the signal 850. The grating coupler 730 can receive at least a second threshold amount (e.g., greater than about thirty percent; between about thirty percent and about ninety percent; between about fifty percent and about eighty percent) of the second component 775 of the return beam 755. For example, the second threshold amount can correspond to an amount required by the grating coupler 730 to output a signal, e.g., the signal 855.

3.3 LIDAR Sensor System Including Beam Shape Optic

[0150] Systems and methods in accordance with the present disclosure can allow for LIDAR sensor systems to more effectively detect objects from which it can be difficult to receive sufficient reflections from transmitted LIDAR beams to achieve target performance criteria. This include, for example, objects that are dark, such as dark vehicles. For example, dark automotive paint on vehicles can have reflections that are highly specular and/or have scattering that is less useful for proper detection. Automotive paint can include, for example, a clear top coat, a translucent layer having high-chroma pigment, and a reflective/absorptive layer having high-brightness small aluminum flakes and/or light-absorbing flakes, applied to a body of a vehicle. The optical complexities of these layers and materials can result in various specular reflections (e.g., from the large change in index of refraction from air to the top coat). Brighter paint colors may depolarize light by acting as volume scatterer; darker paints can maintain polarization as most scatter can be from the top surface and volume scatter can be absorbed. The flakes can cause polarization-maintaining specular reflections. In some instances, the manner in which a LIDAR sensor system's link budget (e.g., configuration of scan rate, beam size, target ranges, spacing between transmit/receive components, etc., to achieve a performance e.g. signal to noise ratio) is tuned can result in lower than expected returns of beams from dark vehicles. Some LIDAR sensors use beams that have a divergence less than an angular spacing from point to point of the beams' scan patterns (e.g., are diffraction limited to have a divergence significantly smaller than the spacing from point to point).

[0151] Any one or more such factors can make it challenging to detect dark vehicles. In an example, a solid black paint may have a reflectivity of 0.1 percent relative to a baseline when five degrees away from the baseline. In various instances, the painted body panels of dark vehicles have very low reflectivity; as a result, many, if not all, detections may have to result from detection of specular or retro reflections and/or any diffuse materials that there may be; for example, due to the beam divergence and scan patterns, specular reflection off of body panels, small retro reflective components such as tail lights or license plates, or small pieces of diffuse materials on tires or trim may fall in between the actual LIDAR beam points and be missed.

[0152] Systems and methods in accordance with the present disclosure can allow for the LIDAR sensor system to more effectively detect objects, including but not limited to dark vehicles, for which reflections may not have expected properties and/or for which the locations of materials having sufficient reflection properties are sparse. For example, one or more optics of the LIDAR sensor system, such as a collimator and/or collimator lens, can be structured to allow for a controlled amount of beam blooming in order to more effectively detect specular reflections. Blooming can correspond to how the LIDAR sensor system determines a representation, such as a point cloud, regarding the objects. Blooming can be when the representation of a given object appears to be larger (e.g., has a greater size in pixel space, such as in pixels of a point cloud) than an actual size of the given object. For example, bright and/or retroreflective objects can have blooming: although much of an transmit beam from the LIDAR sensor system may miss the object, the dim edges of the beam (e.g., where the beam has a Gaussian shape) can reach the portions of the object that cause bright reflection in a manner with sufficient power to cause a detection for points of those portions. Lane marker reflectors, for example, may have a size such that a greater number of detection points result from this blooming effect (e.g., the actual size of a lane marker reflector may be such that only a single LIDAR point should fall on the lane marker reflector). Systems and methods in accordance with the present disclosure can include one or more optics that implement blooming in a manner so that an amount of blooming (e.g., points detected outside of an actual, physical location of an object) does not exceed a target size, such as an angular resolution of the LIDAR sensor system. This can allow the LIDAR sensor system to produce a detection of a point of an object within a region represented by the angular resolution, even if the beam does not directly fall on the point, while avoiding a small retroreflective target from producing detections on multiple points. For example, the LIDAR sensor system can use the optic to spread a small portion of transmit and/or receive light into otherwise empty regions between points to take advantage of bright specular and/or retro-reflective reflections.

[0153] In some implementations, the LIDAR sensor system includes an optic structured to cause a beam outputted by the LIDAR sensor system to have light outside of a target shape for the beam, such as a target Gaussian shape and/or beam divergence. The optic can be a collimation lens, or can be optically coupled with a collimation lens of the LIDAR sensor system. The optic can have an aberration, such as a spherical aberration or a comatic aberration (e.g., coma). The optic can have a birefringent optic to cause different optical path lengths based on polarization, which can allow for a defocus of a portion of the power of the beam to create a larger spot for the beam at the target range (e.g., in the far field). The optic can cause a reduced beam size in one or more directions (e.g., an elliptical beam, such as an elliptical beam having reduced size in the vertical directions), to increase far field spot size. The optic can include a diffractive optical element to achieve the target far field beam shape.

[0154] FIG. 18 depicts an example of a LIDAR sensor system **1800**. The LIDAR sensor system **1800** can incorporate features of various systems and devices described herein, including but not limited to the LIDAR sensor system **500**, the LIDAR sensor system **700**, and/or the optic module **524**. The LIDAR sensor system **1800** can be used to control a shape of beams transmitted by the LIDAR sensor system **1800**, such as to expand or extend, in a controlled manner, where rays of light of the beam are positioned outside of a boundary of a baseline (e.g., target) shape for the beam, such as a diffraction-limited, Gaussian, and/or collimated beam shape. This can allow the transmitted beams to have a greater likelihood of being reflected from a point on the object back towards the LIDAR sensor system **1800** to produce a detection of the point (even if the baseline shape would miss the point, such as to increase probability of detection (POD) of the object).

[0155] As depicted in FIG. 18, the LIDAR sensor system **1800** can include the laser source **504**, which can generate the beam **506**. The LIDAR sensor system **1800** can include optics **532** (e.g., scanning optics), such as a rotatable scanner (e.g., polygon scanner, irregular polygon scanner), to transmit beams **1812** from the LIDAR sensor system **1800** according to a scan pattern. For

example, the optics **532** can be controlled to cause the transmitted beam **1812** to be scanned at one or more points (e.g., first point **1816**, second point **1820**) in an environment around the LIDAR sensor system **1800**, such as one or more points that impinge on object(s) in the environment. The optics **532** can be controlled such that there is a distance **1824** between the points of the scan pattern. The distance **1824** can be a predefined distance, or can be dynamically updated for one or more sweeps of the scan pattern (e.g., by scanner controller **592**). The distance **1824** can represent an angular resolution and/or spatial resolution of the LIDAR sensor system **1800** while performing the scan according to the scan pattern. For example, the distance **1824** can correspond to an angle between the first point **1816** and the second point **1820**, such as between respective centers of beams targeted to the first point **1816** and the second point **1820**. In some implementations, the angular resolution is between about 0.05 degrees and 0.5 degrees. In some implementations, the angular resolution includes a first component (e.g., azimuth component) and a second component (e.g., elevation component) different from the first component, such as to scan the beam more tightly in azimuth relative to elevation, or vice versa.

[0156] The beam **506** can have a beam shape. The beam shape can represent a location of rays of light of the beam **506**, such as relative to a centerline or axis (e.g., optical axis) of the beam as received at the optics **532** and/or outputted from the laser source **504**. The beam shape can be defined in a plane perpendicular to the axis of the beam, such as to be a cross-sectional shape of the beam. In some implementations, the beam shape is defined based on a distribution of light and/or power of the beam in the plane relative to the axis (e.g., radially outward from the axis).

[0157] The LIDAR sensor system **1800** (e.g., laser source **504** and/or optics module **524**) can generate the beam **506** to have a target shape for the beam shape. The target shape can be at least one of a collimated shape, a Gaussian shape, or a diffraction-limited beam shape. The Gaussian shape can be defined based on the distribution of the light and/or power of the beam being within a threshold (e.g., a maximum threshold, such as less than or equal to about ten percent; less than or equal to about five percent; between about 0.1 percent and about ten percent) of a Gaussian (e.g., normal) distribution.

[0158] In some implementations, the target shape is such that at least a threshold amount of the beam **506** (e.g., power of the light of the beam) is within a boundary around the centerline (e.g., at a given distance along the path of the beam **506**), such as to be no further radially outward from the centerline than the boundary. The threshold amount can be, for example, greater than or equal to about 68 percent; greater than or equal to about 95 percent; greater than or equal to about 99 percent; or various other higher thresholds less than 100 percent. In some implementations, the target shape corresponds to a beam quality factor, such as a maximum value of the M^2 factor comparing the shape and/or divergence of the beam to a Gaussian beam, where values greater than 1 correspond to deviations from a Gaussian beam:

[00001] $M^2 = \frac{\omega_{sub.0}}{\lambda}$ [0159] where $\omega_{sub.0}$ is a beam waist of the beam **506**, θ is a divergence angle of the beam **506**, and λ is a wavelength of the beam **506**. For example, in some implementations, the M^2 factor for the target shape is greater than 1 and less than or equal to about []].

[0160] The LIDAR sensor system **1800** can include at least one optic **1804**, which can be arranged on an optical path between the laser source **504** and the optics **532**. In some implementations, the LIDAR sensor system **1800** includes any of various optical components between the laser source **504** and the optic **1804** and/or between the optic **1804** and the optics **532**, including but not limited to the splitter **508**, modulator **514**, amplifier **520**, and/or optic module **524** or one or more components thereof. For example, various operations can be performed on the beam **506** from the laser source **504** to the optic **1804** prior to reception of the beam **506** by the optic **1804**.

[0161] The optic **1804** can modify the beam shape of the beam **506** to output a beam **1808**, such as to provide for a greater amount of the light and/or power of the beam **506** to be outside of the boundary of the target shape for the beam **506**. For example, the optic **1804** can modify the beam shape such that more of the light and/or power of the beam **506** is outside of the boundary

(represented by the threshold amount, the divergence, and/or the M.sup.2 factor) upon output of the beam **506** from the optic **1804** relative to input of the beam **506** to the optic **1804**. In some implementations, the optic **1804** modifies the beam shape of the beam **506** so that the beam **1808** includes light outside of the boundary and within the distance from the center of the beam **1808**.

For example,

[0162] The optic **1804** can be or include a lens having an aberration. The aberration can be caused by at least one of a structure of the optic **1804** or a pattern of the optic **1804**, such as a pattern formed on the optic **1804**. The aberration can be a spherical aberration, such as where rays further outward from the optical axis of the optic **1804** have a focal point closer to the optic **1804** relative to rays closer to the optical axis. The aberration can be a comatic aberration, such as to have light directed off-center in a portion of a field of view of the optic **1804**. In some implementations, the aberration of the optic **1804** can be represented by one or more Zernike polynomials; for example, the optic **1804** can be structured (e.g., have structures formed in and/or patterns formed on) in a manner represented by a Zernike polynomial.

[0163] In some implementations, the optic **1804** can modify the beam **506** so that the beam **1808** has an elliptical shape. For example, the beam **1808** can extend further in a first direction perpendicular to the axis of the beam **1808** than a second direction perpendicular to the first direction and perpendicular to the axis. In some implementations, the elliptical shape is such that the beam **1808** has a reduced vertical size (e.g., the first direction is in azimuth, and the second direction is in elevation).

[0164] In some implementations, the optic **1804** includes a diffractive element, which can diffract a portion of the beam **506** to modify the beam shape of the beam **506**. For example, the diffractive element can be a diffraction grating. The use of the diffractive element can allow for precise shaping of the beam **1808**.

[0165] In some implementations, the optic **1804** includes or is coupled with the optic module **524**, such as to include or be coupled with one or more collimation lenses of the optic module **524**. For example, the optic **1804** can be a collimation lens, or the one or more collimation lenses of the optic module **524** can be arranged on the optical path from the laser module **504** to the optic **1804** to the optics **532**. For example, the one or more collimation lenses can collimate the beam **506** and provide the collimated beam **506** to the optic **1804**, and the optic **1804** can modify the collimated beam **506** to adjust the collimated beam **506** to have a greater amount of light outside of the beam shape of the collimated beam **506** to output as the beam **1808**.

[0166] Having now described some illustrative implementations, it is apparent that the foregoing is illustrative and not limiting, having been presented by way of example. In particular, although many of the examples presented herein involve specific combinations of method acts or system elements, those acts and those elements can be combined in other ways to accomplish the same objectives. Acts, elements and features discussed in connection with one implementation are not intended to be excluded from a similar role in other implementations or implementations.

[0167] The phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of “including” “comprising” “having” “containing” “involving” “characterized by” “characterized in that” and variations thereof herein, is meant to encompass the items listed thereafter, equivalents thereof, and additional items, as well as alternate implementations consisting of the items listed thereafter exclusively. In one implementation, the systems and methods described herein consist of one, each combination of more than one, or all of the described elements, acts, or components.

[0168] Any references to implementations or elements or acts of the systems and methods herein referred to in the singular can also embrace implementations including a plurality of these elements, and any references in plural to any implementation or element or act herein can also embrace implementations including only a single element. References in the singular or plural form are not intended to limit the presently disclosed systems or methods, their components, acts, or

elements to single or plural configurations. References to any act or element being based on any information, act or element can include implementations where the act or element is based at least in part on any information, act, or element.

[0169] Any implementation disclosed herein can be combined with any other implementation or embodiment, and references to “an implementation,” “some implementations,” “one implementation” or the like are not necessarily mutually exclusive and are intended to indicate that a particular feature, structure, or characteristic described in connection with the implementation can be included in at least one implementation or embodiment. Such terms as used herein are not necessarily all referring to the same implementation. Any implementation can be combined with any other implementation, inclusively or exclusively, in any manner consistent with the aspects and implementations disclosed herein.

[0170] Where technical features in the drawings, detailed description or any claim are followed by reference signs, the reference signs have been included to increase the intelligibility of the drawings, detailed description, and claims. Accordingly, neither the reference signs nor their absence have any limiting effect on the scope of any claim elements.

[0171] Systems and methods described herein may be embodied in other specific forms without departing from the characteristics thereof. Further relative parallel, perpendicular, vertical or other positioning or orientation descriptions include variations within $\pm 10\%$ or ± 10 degrees of pure vertical, parallel or perpendicular positioning. References to “approximately,” “about” “substantially” or other terms of degree include variations of $\pm 10\%$ from the given measurement, unit, or range unless explicitly indicated otherwise. Coupled elements can be electrically, mechanically, or physically coupled with one another directly or with intervening elements. Scope of the systems and methods described herein is thus indicated by the appended claims, rather than the foregoing description, and changes that come within the meaning and range of equivalency of the claims are embraced therein.

[0172] The term “coupled” and variations thereof includes the joining of two members directly or indirectly to one another. Such joining may be stationary (e.g., permanent or fixed) or moveable (e.g., removable or releasable). Such joining may be achieved with the two members coupled directly with or to each other, with the two members coupled with each other using a separate intervening member and any additional intermediate members coupled with one another, or with the two members coupled with each other using an intervening member that is integrally formed as a single unitary body with one of the two members. If “coupled” or variations thereof are modified by an additional term (e.g., directly coupled), the generic definition of “coupled” provided above is modified by the plain language meaning of the additional term (e.g., “directly coupled” means the joining of two members without any separate intervening member), resulting in a narrower definition than the generic definition of “coupled” provided above. Such coupling may be mechanical, electrical, or fluidic.

[0173] References to “or” can be construed as inclusive so that any terms described using “or” can indicate any of a single, more than one, and all of the described terms. A reference to “at least one of ‘A’ and ‘B’” can include only ‘A’, only ‘B’, as well as both ‘A’ and ‘B’. Such references used in conjunction with “comprising” or other open terminology can include additional items.

[0174] Modifications of described elements and acts such as variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations can occur without materially departing from the teachings and advantages of the subject matter disclosed herein. For example, elements shown as integrally formed can be constructed of multiple parts or elements, the position of elements can be reversed or otherwise varied, and the nature or number of discrete elements or positions can be altered or varied. Other substitutions, modifications, changes and omissions can also be made in the design, operating conditions and arrangement of the disclosed elements and operations without departing from the scope of the present disclosure.

[0175] References herein to the positions of elements (e.g., “top,” “bottom,” “above,” “below”) are merely used to describe the orientation of various elements in the FIGURES. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure.

Claims

1. A light detection and ranging (LIDAR) sensor system for a vehicle, the LIDAR sensor system comprising: a laser source configured to generate a beam; one or more optics configured to provide a modified beam by modifying a cross-sectional shape of the beam to be outside of a boundary of a target shape for the beam; and one or more scanners configured to output the modified beam.
2. The LIDAR sensor system of claim 1, wherein: the cross-sectional shape of the beam is defined in a plane perpendicular to an axis of the beam; the target shape is a Gaussian shape such that a distribution of a power of the beam outward from the axis is within a threshold of a Gaussian distribution; the one or more scanners are configured to output the modified beam in a scan pattern having a first point at which the modified beam is outputted and a second point at which the modified beam is outputted such that there is a predefined distance between the first point and the second point; and the one or more optics are configured to modify the cross-sectional shape to include light outside of the boundary and within the predefined distance from a center of the modified beam such that the M.sup.2 factor for the modified beam is greater than 1 and less than or equal to about []].
3. The LIDAR sensor system of claim 1, wherein the one or more optics comprise a lens having an aberration to modify the cross-sectional shape.
4. The LIDAR sensor system of claim 3, wherein the aberration comprises at least one of a spherical aberration or a comatic aberration.
5. The LIDAR sensor system of claim 1, wherein the one or more optics are structured according to a Zernike polynomial that represents the cross-sectional shape of the modified beam.
6. The LIDAR sensor system of claim 1, wherein the one or more optics comprise a birefringent optic configured to defocus a portion of the beam to modify the cross-sectional shape.
7. The LIDAR sensor system of claim 1, wherein the one or more optics are configured to modify the beam to have an elliptical shape.
8. The LIDAR sensor system of claim 1, wherein the one or more optics comprise a diffractive element configured to diffract a portion of the beam to modify the cross-sectional shape.
9. The LIDAR sensor system of claim 1, further comprising a collimator coupled with the laser source to collimate the beam to have the target beam shape, wherein the one or more optics are configured to receive the beam from the collimator as a collimated beam and modify the cross-sectional shape of the beam by modifying the collimated beam.
10. An autonomous vehicle control system, comprising: a laser source configured to generate a beam; one or more optics configured to provide a modified beam by modifying a cross-sectional shape of the beam to be outside of a boundary of a target shape for the beam; one or more scanners configured to output the modified beam; and one or more processors configured to: determine at least one of a range to an object or a velocity of the object based on a return signal from reflection of the modified beam by the object; and control operation of an autonomous vehicle responsive to the at least one of the range or the velocity.
11. The autonomous vehicle control system of claim 10, wherein: the target shape is a Gaussian shape; the vehicle controller is configured to control the one or more scanners to output the modified beam in a scan pattern having a first point at which the modified beam is outputted and a second point at which the modified beam is outputted such that there is a predefined distance between the first point and the second point; and the one or more optics are configured to modify the cross-sectional shape to include light outside of the boundary and within the predefined

distance from a center of the modified beam.

- 12.** The autonomous vehicle control system of claim 10, wherein the one or more optics comprise a lens having an aberration to modify the cross-sectional shape.
 - 13.** The autonomous vehicle control system of claim 10, wherein the one or more optics are structured according to a Zernike polynomial that represents the cross-sectional shape of the modified beam.
 - 14.** The autonomous vehicle control system of claim 10, wherein the one or more optics comprise a birefringent optic configured to defocus a portion of the beam to modify the cross-sectional shape.
 - 15.** The autonomous vehicle control system of claim 10, wherein the one or more optics are configured to modify the beam to have an elliptical shape.
 - 16.** The autonomous vehicle control system of claim 10, wherein the one or more optics comprise a diffractive element configured to diffract a portion of the beam to modify the cross-sectional shape.
 - 17.** The autonomous vehicle control system of claim 10, further comprising a collimator coupled with the laser source to collimate the beam to have the target beam shape, wherein the one or more optics are configured to receive the beam from the collimator as a collimated beam and modify the cross-sectional shape of the beam by modifying the collimated beam.
 - 18.** An autonomous vehicle, comprising: a LIDAR sensor system, comprising: a laser source configured to generate a beam; one or more optics configured to provide a modified beam by modifying a cross-sectional shape of the beam to be outside of a boundary of a target shape for the beam; and one or more scanners configured to output the modified beam; a steering system; a braking system; and a vehicle controller comprising one or more processors configured to: determine at least one of a range to an object or a velocity of the object based on a return signal from reflection of the modified beam by the object; and control operation of at least one of the steering system and the braking system responsive to the at least one of the range or the velocity.
 - 19.** The autonomous vehicle of claim 18, wherein: the target shape is a Gaussian shape; the one or more scanners are configured to output the modified beam in a scan pattern having a first point at which the modified beam is outputted and a second point at which the modified beam is outputted such that there is a predefined distance between the first point and the second point; and
 - 20.** The autonomous vehicle of claim 18, further comprising a collimator coupled with the laser source to collimate the beam to have the target beam shape, wherein the one or more optics are configured to receive the beam from the collimator as a collimated beam and modify the cross-sectional shape of the beam by modifying the collimated beam.
-